Pure Mathematics with Statistics GCE A Level 2023 - Paper 1

Duration: 1h30 | Cameroon GCE Board

Multiple choice paper (0770 Pure Maths With Statistics 1, June 2023, Advanced Level, one and a half hours). Answer ALL questions: Section A is Pure Mathematics (questions 1 to 35) and Section B is Statistics (questions 36 to 50). Formulae booklets and calculators are allowed. Each question has FOUR suggested answers A, B, C and D; mark only one answer for each question (if you mark more than one answer you score zero). Avoid spending too much time on any one question. Mobile phones are NOT allowed in the examination room.

SECTION A: PURE MATHEMATICS

Given that $4^{3x+2} = 2^{6+2x}$, the value of x is

a) $-\frac{1}{2}$
b) 2
c) $\frac{1}{2}$
d) 1

2. Two functions f and g are defined on the set of real numbers $\mathbb{R}$ by $f(x) = 2 - 3x$ and $g(x) = \frac{4x-1}{2}$, $g \circ f(-1) =$

Options as printed: A) $\frac{19}{3}$ B) $\frac{19}{3}$ C) $\frac{7}{2}$ D) -3

3. Two statements p and q are logically equivalent if

a) p and q are always false
b) p and q are linked by a logic operator
c) p and q have identical truth tables
d) p is the converse of q

4. If $\frac{x-5}{1-x^2} \equiv \frac{M}{1+x} + \frac{N}{1-x}$, then

a) $M = -3$ and $N = 2$
b) $M = -3$ and $N = -2$
c) $M = 3$ and $N = -2$
d) $M = 3$ and $N = 2$

5. The value of t for which the plane $2x + 4y + tz = 9$ and the line $\mathbf{r} = 2\mathbf{i} + 4\mathbf{j} + 9\mathbf{k} + \lambda(3\mathbf{i} - 5\mathbf{j} + 2\mathbf{k})$ are parallel is

a) 7
b) 6
c) -3
d) 2

6. The Cartesian equation of the curve with parametric equation $x = 1 + \sin\theta, y = 2 - \cos\theta$ is

a) $x^2 + y^2 + 2x + 4y + 4 = 0$
b) $x^2 + y^2 - 2x - 4y + 4 = 0$
c) $x^2 + y^2 - 2x - 4y - 4 = 0$
d) $x^2 + y^2 - 2x + 4y - 4 = 0$

7. The third term in the binomial expansion of $(1 - 4x)^{-6}$ is

a) $-336x^2$
b) $336x^3$
c) $336x^2$
d) $-336x^3$

8. The general solution of the differential equation $(1 + x^2)\dfrac{dy}{dx} = 2$ is

a) $y = 2\tan^{-1} x + A$
b) $y = 2\tan^{-1} \dfrac{x}{2} + A$
c) $y = \tan^{-1} 2x + A$
d) $y = \tan^{-1} \dfrac{x}{2} + A$

9. All the asymptotes of the graph of the function $f(x) = \dfrac{5x^2 + 3}{(x + 2)(x - 1)}$ are

a) $x = -1,\ x = 2,\ y = -5$
b) $x = -1,\ x = -2,\ y = 5$
c) $x = 1,\ x = 2,\ y = -5$
d) $x = 1,\ x = -2,\ y = 5$

10. The value of m for which $\begin{vmatrix} 3 & m & -1 \\ 1 & -1 & 3 \\ 2 & 0 & 2 \end{vmatrix} = -2$ is

a) $\dfrac{2}{3}$
b) $\dfrac{1}{3}$
c) $\dfrac{3}{2}$
d) $\dfrac{3}{4}$

11. The range of values of x for which $(3 - x)(7 + x) \leq 0$ is

a) $x \leq -7$ or $x \geq 3$
b) $x < -7$ or $x \geq 3$
c) $x \leq -3$ or $x \geq 7$
d) $-7 \leq x \leq 3$

12. A function f defined by $f(x) = \begin{cases} 2x, & 0 \leq x \leq 2, \\ 3x^2 + 1, & 2 \leq x \leq 3, \end{cases}$ is periodic with period 3. $f(9) =$

a) 0
b) 1
c) 4
d) 2

13. The remainder when $x^3 - 3x^2 - 4x - 12$ is divided by $x + 2$ is

a) 16
b) -24
c) 24
d) -16

14. The eighth term of a geometric progression is 256 and the first term is 2. The common ratio is

a) 7
b) 128
c) 2
d) 18

15. The number of arrangements of the letters of the word $ARREARS$ is

a) 210
b) 420
c) 840
d) 640

16. The Cartesian equation of the plane containing the point $\mathbf{i} + 3\mathbf{j} - 8\mathbf{k}$ and perpendicular to the line $\mathbf{r} = 3\mathbf{i} + 7\mathbf{j} - 2\mathbf{k} + t(5\mathbf{i} - \mathbf{j} - 4\mathbf{k})$ is

a) $5x - y + 4z = 34$
b) $5x + y - 4z = 34$
c) $-5x + y - 4z = 34$
d) $5x - y - 4z = 34$

17. Given the matrix $\begin{pmatrix} 2 & 1 & 5 \\ -1 & 2 & 3 \\ 4 & 1 & 7 \end{pmatrix}$, the cofactor of 7 is

a) 3
b) 4
c) 5
d) -3

18. $\log 24 - \frac{1}{2}\log 9 + \log 125 =$

a) 3
b) 2
c) 10
d) 4

19. Given that $\sin\theta = \frac{1}{2}$, the values of θ, where $0 < \theta < \pi$, are

a) $\frac{\pi}{6}, \frac{5\pi}{6}$
b) $\frac{\pi}{2}, \frac{2\pi}{3}$
c) $\frac{\pi}{3}, \frac{2\pi}{3}$
d) $\frac{\pi}{3}, \frac{3\pi}{4}$

20. The vector equations of 4 straight lines L_1, L_2, L_3, L_4 are given by: $L_1: \mathbf{r} = 3\mathbf{i} - 2\mathbf{j} + 4\mathbf{k} + \lambda_1(4\mathbf{i} + 2\mathbf{j} - 3\mathbf{k})$, $L_2: \mathbf{r} = \mathbf{i} + 3\mathbf{j} + \mathbf{k} + \lambda_2(8\mathbf{i} + 4\mathbf{j} - 6\mathbf{k})$, $L_3: \mathbf{r} = 2\mathbf{i} + 6\mathbf{j} + 2\mathbf{k} + \lambda_3(\mathbf{i} - 4\mathbf{j} + 4\mathbf{k})$, $L_4: \mathbf{r} = 4\mathbf{i} + 12\mathbf{j} + 4\mathbf{k} + \lambda_4(3\mathbf{i} - \mathbf{j} + \mathbf{k})$. The parallel pair of lines is

a) L_1 and L_2
b) L_1 and L_4
c) L_2 and L_4
d) L_3 and L_4

21. A function f is defined by $f(x) = \begin{cases} 4\beta + 4x & \text{if } x < 1, \\ 4x - \beta^2 - 4 & \text{if } x > 1. \end{cases}$ If f has a limit as $x \to 1$, the value of the constant β is

a) 2
b) 1
c) 3
d) -2

22. $\frac{2 + i - i^4}{i + i^2 + i^4} =$

a) $1 - i$
b) $1 + i$
c) $-1 + i$
d) $-1 - i$

23. A relation R defined on a non-empty set S is called a partial order if R is

a) Reflexive, Symmetric and Anti-symmetric
b) Reflexive, Anti-symmetric and Transitive
c) Reflexive, Symmetric and transitive
d) Transitive, Symmetric and Anti-symmetric

24. $(\mathbf{i} + 2\mathbf{j} + 2\mathbf{k}) \times (-2\mathbf{i} + 3\mathbf{j} + \mathbf{k}) =$

a) $-4\mathbf{i} - 5\mathbf{j} - 7\mathbf{k}$
b) $-4\mathbf{i} + 5\mathbf{j} + 7\mathbf{k}$
c) $-4\mathbf{i} - 5\mathbf{j} + 7\mathbf{k}$
d) $-4\mathbf{i} + 5\mathbf{j} - 7\mathbf{k}$

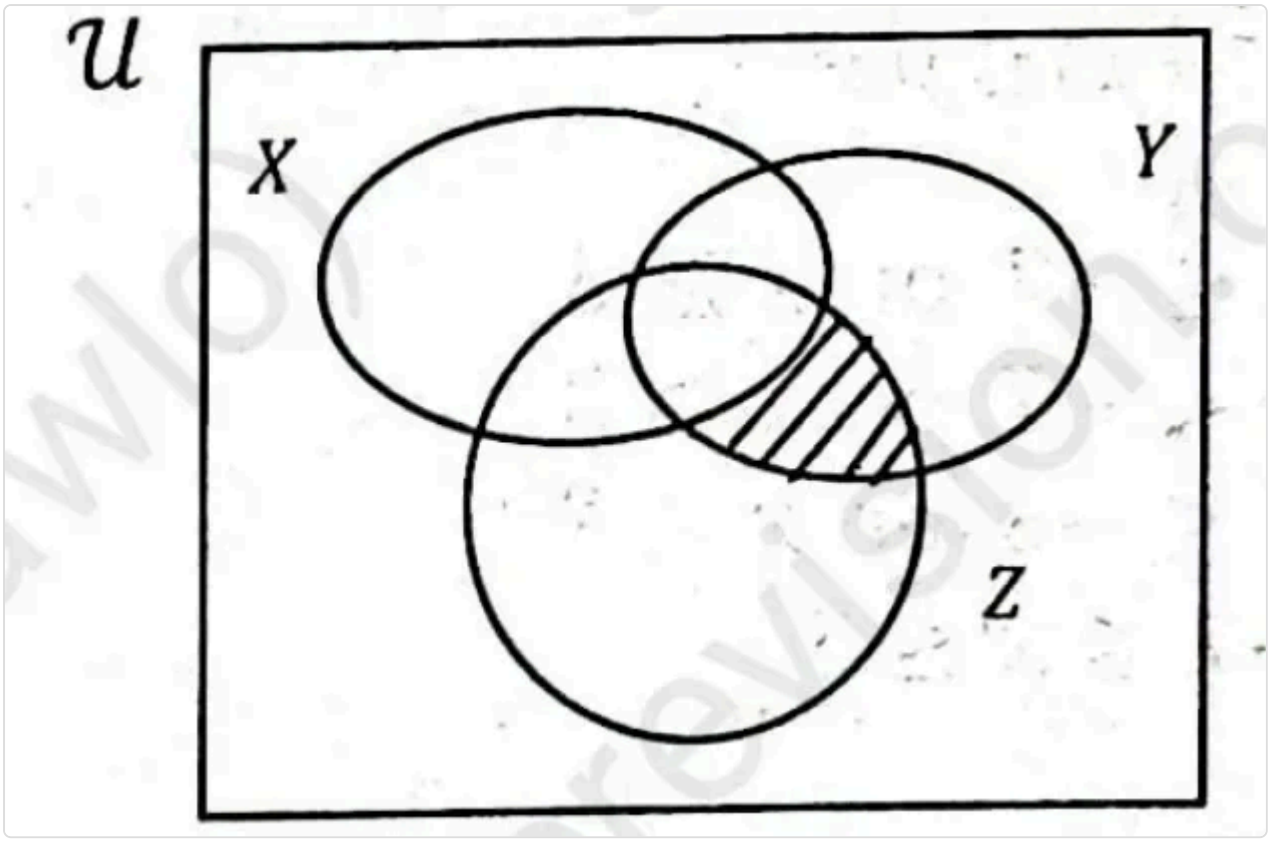
25. Given that $\sin\theta - \sqrt{3}\cos\theta = R\sin(\theta - \lambda)$, where $R > 0$ and λ is an acute angle, R and λ are such that

a) $R = 2$ and $\lambda = 90^\circ$
b) $R = \sqrt{10}$ and $\lambda = 60^\circ$
c) $R = 2$ and $\lambda = 60^\circ$
d) $R = \sqrt{10}$ and $\lambda = 45^\circ$

26. The roots of the equation $2x^2 - 3x - 1 = 0$ are α and β. The equation whose roots are $2\alpha + 1$ and $2\beta + 1$ is

a) $x^2 + 5x - 2 = 0$
b) $x^2 + 5x + 2 = 0$
c) $x^2 - 5x - 2 = 0$
d) $x^2 - 5x + 2 = 0$

27.



The set represented by the shaded region is

a) $X' \cap Y \cap Z$
b) $Y \cap Z$
c) $(Y \cap Z) \cup (X \cap Y \cap Z)$
d) $(Y' \cap Z') - X$

28. The relationship between the logarithms of the variables x and y which are connected by $2y + 10 = ab^{(x-3)}$ is

a) $\log(2y + 10) = (x - 3)\log b - \log a$
b) $\log(2y + 10) = (x - 3)\log b + \log a$
c) $\log(2y + 10) = (x - 3)(\log b + \log a)$
d) $\log(2y + 10) = (x - 3)\log a - \log b$

29. The sum of the first n terms of an arithmetic progression is S_n, where $S_n = n(2n + 3)$. The nth term of the progression is

a) $6n + 1$
b) $2n + 1$
c) $4n - 1$
d) $4n + 1$

30. The values of y corresponding to values of x in an experiment are given on the table below.

x	**0**	**2**	**4**	**6**	**8**	**10**
y	0.3	0.8	1.6	2.4	2.7	5.3

Using the trapezium rule, the approximate value of $\int_0^{10} y\,dx$ is

Options as printed: A) 19.2 B) 11.3 C) 12.1 D) 18.8

31. The first approximation to the root of the equation $x^3 - 4 = 0$ is 1.5. Using the Newton-Raphson method, the second approximation to the root of the equation is

a) 1.594
b) 1.593
c) 1.592
d) 1.595

32. The greatest value of $5\sin\theta - 12\cos\theta + 20$ is

a) 33
b) 7
c) 13
d) -13

33. Given that $z = \sqrt{3} + i$, $\arg(z^6) =$

a) $\frac{\pi}{3}$
b) π
c) $\frac{\pi}{6}$
d) $\frac{5\pi}{6}$

34. $\int 2x(3x^2 + 5)^7\,dx =$

a) $\frac{8}{3}(3x^2 + 5)^8 + C$
b) $\frac{1}{24}(3x^2 + 5)^8 + C$
c) $\frac{16x}{3}(3x^2 + 5)^8 + C$
d) $\frac{x}{12}(3x^2 + 5)^8 + C$

35. If $f : \mathbb{R} \to \mathbb{R}$ is given by $f(x) = \sqrt{\frac{2}{x-2} - 3}$, the domain of f is

a) $\mathbb{R} - \{2\}$
b) $2 \leq x \leq \frac{8}{3}$
c) \$2 d) $x < \frac{8}{3},\ x \neq 2$

SECTION B: STATISTICS

The median of the set of data $3, 4, 4, 6, 7, 8, 9, 3, 4, 7, 8, 9, 3, 8, 9, 4$ is

a) 4.0
b) 6.0
c) 6.5
d) 8.5

37. Two independent events A and B are such that $P(A) = 0.3$ and $P(A \cup B) = 0.58$. $P(B) =$

a) 0.40
b) 0.60
c) 0.42
d) 0.88

38. The set of ordered pairs $\{(x, y)\}$, $\{(4, 8), (5, 9), (6, 6), (7, 3)\}$ is such that $\bar{x} = 5.50$ and $\bar{y} = 6.50$. The covariance of the ordered pairs is

a) 2.25
b) 1.50
c) -1.50
d) -2.25

39. A random variable X has cumulative distribution function $F(x) = k(3x - 1)$ for $x = 1, 2, 3$. The value of the real constant k is

a) $\frac{1}{2}$
b) $\frac{1}{5}$
c) $\frac{1}{8}$
d) $\frac{1}{15}$

40. Random variable $X \sim B(n, p)$. If $E(X) = 2$ and $\text{Var}(X) = 1.6$, the value of n is

a) 25
b) 32
c) 8
d) 10

41. A frequency distribution is such that: Lower quartile $= 22.5$, Median $= 24.3$ and Upper quartile $= 28.7$. The quartile deviation for the distribution is

a) 3.1
b) 6.2

c) 1.8
d) 2.2

42. A sample of size 30 is drawn from an infinite population with mean μ and variance σ^2. The variance of the sample mean is

a) $\frac{\sigma}{29}$
b) $\frac{\sigma}{30}$
c) $\frac{\sigma^2}{29}$
d) $\frac{\sigma^2}{30}$

43. A sample of size 8 drawn from a population gives rise to $\sum x = 391$, $\sum x^2 = 19159$. Using this sample, the unbiased estimate of the population variance is

Options as printed: none (the source paper prints no answer options for this question).

44. A population has mean 15 and standard deviation 2.3. A random sample of size 81 is taken from the population and its mean is found to be 15.4. A hypothesis test is to be conducted to verify if the mean has changed. The value of the test statistic for this test is

a) -1.645
b) -1.565
c) 1.565
d) 1.645

45. For ten pairs of values (x, y), $\bar{x} = 13$, $\bar{y} = 7.08$, $S_x^2 = 1.246$, $S_y^2 = 1.611$ and $S_{xy} = 1.389$. The product moment correlation coefficient is

a) 0.98
b) 0.69
c) 0.89
d) 0.83

46. A continuous random variable X has probability density function f given by $f(x) = \frac{3}{16}x(4 - x)$, $2 \leq x \leq 4$. The mode of X is

a) 2.4
b) 2.2
c) 2.5
d) 2.0

47. A continuous random variable X has probability density function g given by $g(x) = ax(2 - x)$, $0 \leq x \leq 2$. The value of the constant a is

a) $\frac{5}{8}$
b) $\frac{3}{4}$

c) $\frac{3}{8}$
d) $\frac{1}{4}$

48. A random variable $X \sim \text{Po}(\lambda)$. If $49P(X=0) = 8P(X=2)$, the value of the parameter λ is

a) 12.25
b) 3.50
c) 3.00
d) 1.75

49. For two events A and B, $P(A) = \frac{1}{5}$, $P(B) = \frac{3}{5}$, $P(B \mid A) = \frac{1}{10}$. The value of $P(A \cap B)$ is

a) $\frac{3}{50}$
b) $\frac{1}{50}$
c) $\frac{3}{25}$
d) $\frac{9}{10}$

50. A discrete random variable $X \sim B(100, 0.2)$. Using the normal approximation, $P(X > 30)$ is equivalent to

Options as printed: A) $P(Z > 1.88)$ B) $P(Z > 2.52)$ C) $P(Z > 1.86)$ D) $P(Z > 2.50)$

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